

EESC4550/5550: Practical Atmospheric Modeling (Fall 2026)

Lab 2 – Modifying and Compiling the Chemical Kinetic Preprocessor

Total Score: 10

Learning Objectives:

- Add dummy tracer species to a GEOS-Chem KPP mechanism to track production pathways of nitrate species.
- Update `species_database.yml` consistently with new KPP species and rebuild the KPP mechanism by recompilation.

Modifying Chemical Mechanisms in KPP

First, navigate to `/lustre/project/eesc4550/<your-computing-id>/my_code_dir/run` to create a new run directory under `/lustre/project/eesc4550/<your-computing-id>/GEOS-Chem/GC_rundirs/`. Name this new run directory as `KPP`.

The KPP code module is located at `/lustre/project/eesc4550/<your-computing-id>/GEOS-Chem/GC_rundirs/KPP/CodeDir/src/GEOS-Chem/KPP/fullchem/`.

1. Adding HNO₃ Production Tracking Species

fullchem.kpp

Name species as families for production metric for HNO₃, by adding them to the `#FAMILIES` section.

```
#FAMILIES          { Chemical families for prod/loss diagnostic }  
  
...  
PH2O2 : H2O2;  
PHNO3: HNO3;  
PHNO3NO2OH: HNO3NO2OH;  
PHNO3N2O5H2O: HNO3N2O5H2O;  
PHNO3other: HNO3other;
```

fullchem.eqn

```
...  
#DEFVAR  
  
...  
ZR02      = IGNORE; {Lumped R02 from aromatics}  
HNO3NO2OH  = IGNORE; {Dummy; NO2+OH}  
HNO3N2O5H2O = IGNORE; {Dummy; N2O5+H2O}  
HNO3other  = IGNORE; {Dummy; other}  
#DEFFIX
```

2. Modifying Chemical Equations to Track HNO₃ Production

fullchem.eqn

#EQUATIONS

// %%

// %%%% Reactions extracted from sulfate_mod.F90 (MSL, BMY) %%%%

// %%

//

...

NO2 + OH {+M} = HNO3 + HNO3NO2OH {+M} : GCJPLPR_aba(1.80d-30, 3.0d+00, 2.8d-11, 0.6d0);

...

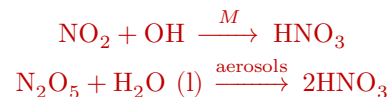
NO3 + CH2O = HNO3 + H2O + CO + HNO3other : 5.80d-16;

...

N2O5 = 2.000HNO3 + 2.000HNO3N2O5H2O : N2O5uptkByCloud(State_Het);

{2018/10/17; Cloud uptake, CDH}

...



There should be **46** chemical equations with HNO₃ as product, update each equation by adding the correct tracer species with the correct coefficient in front of it. That is, if 2 mols of HNO₃ are generated as a product, the coefficient in front of the tracer must also be 2.

Hint: if you are on VS Code, you may use the find function with regex `=.*HNO3` for all the lines containing the equations that we care about. Here, represents the space character; if you are on vim, type the expression `/=.*HNO3\s` under the normal mode.

3. Update Entries in `species_database.yml`

```
# GEOS-Chem Species Database
# Core species only (neglecting microphysics)
# NOTE: Anchors must be defined before any variables that reference them.
...
XYLE:
  Formula: C8H10
  FullName: Xylene
  Is_Advected: true
  Is_Gas: true
  MW_g: 106.18
HNO3NO2OH:
  FullName: Dummy species to track production of HNO3
  Is_Gas: true
  MW_g: 63.012
HNO3N2O5H2O:
  FullName: Dummy species to track production of HNO3
  Is_Gas: true
  MW_g: 63.012
HNO3other:
  FullName: Dummy species to track production of HNO3
  Is_Gas: true
  MW_g: 63.012
PHNO3:
  FullName: Dummy species to track production of HNO3
  Is_Gas: true
  MW_g: 63.012
PHNO3NO2OH:
  FullName: Dummy species to track production of HNO3 from NO2 + OH path
  Is_Gas: true
  MW_g: 63.012
PHNO3N2O5H2O:
  FullName: Dummy species to track production of HNO3 from N2O5 + H2O path
  Is_Gas: true
  MW_g: 63.012
PHNO3other:
  FullName: Dummy species to track production of HNO3 from other path
  Is_Gas: true
  MW_g: 63.012
```

4. Recompile KPP and GEOS-Chem

```
ssh sandbox
source /lustre/project/eesc4550/GEOS-Chem/scripts/gcc.env
cd /lustre/project/eesc4550/<your-computing-id>/GEOS-Chem/GC_rundirs/KPP/CodeDir/src/GEOS-Chem/KPP
./build_mechanism.sh fullchem >& log.kpp
```

Make sure that there is no error message in `log.kpp`.

```
cd /lustre/project/eesc4550/<your-computing-id>/GEOS-Chem/GC_rundirs/KPP/build
make clean
cmake ../CodeDir -DRUNDIR=..
make -j
make -j install
```

5. Add Output Diagnostics

HISTORY.rc

```
COLLECTIONS: 'Restart',
             #'Metrics',
             ...
             'ProdLoss',
             #'BoundaryConditions',
```

For this lab's purpose, change `ProdLoss.frequency` to generate an output per one hour.

geoschem_config.yml

Then, you may run the GEOS-Chem simulation, use `ncdump` to show that you have successfully generated the three HNO_3 production rates corresponding to the three pathways.

```
#=====
# Simulation settings
#=====
simulation:
  name: fullchem
  start_date: [20190701, 000000]
  end_date: [20190701, 010000]      # you will change Restart.frequency
  ...
```

Submission Guideline: To complete the lab, please submit **THREE** screenshots:

- `lab2_<your_id>_{kpp,gc1,gc2}.{png,jpg,jpeg}`, which contains the screenshots (one from `log.kpp` and two from GEOS-Chem: running log and variable existence check) demonstrating that you have successfully completed the re-compilation.

----- End of Lab 2 -----

Appendix: Screenshot Examples

```
[b169195@chpc-sandbox KPP]$ cat log.kpp
```

This is KPP-3.5.0.

```
KPP is parsing the equation file.,,
KPP is computing Jacobian sparsity structure.
KPP is starting the code generation.
KPP is initializing the code generation.
KPP is generating the monitor data:
- gckpp_Monitor
KPP is generating the utility data:
- gckpp_Util
KPP is generating the global declarations:
- gckpp_Main
KPP is generating the ODE function:
- gckpp_Function
KPP is generating the ODE Jacobian:
- gckpp_Jacobian
- gckpp_JacobianSP
KPP is generating the linear algebra routines:
- gckpp_LinearAlgebra
KPP is generating the utility functions:
- gckpp_Util
KPP is generating the rate laws:
- gckpp_Rates
KPP is generating the parameters:
- gckpp_Parameters
KPP is generating the global data:
- gckpp_Global
KPP is generating the driver from none.F90:
- gckpp_Main
KPP is starting the code post-processing.

KPP has successfully created the model "gckpp".
```

Reactivity consists of ██████ reactions

Written to gckpp Util.F90

```
--> DATE: 2019/07/01 UTC: 00:10
--> DATE: 2019/07/01 UTC: 00:20
--> DATE: 2019/07/01 UTC: 00:30
--> DATE: 2019/07/01 UTC: 00:40
--> DATE: 2019/07/01 UTC: 00:50
- Creating file for ProdLoss; reference = 20190701_000000
  with filename = OutputDir/GEOSChem.ProdLoss.20190701_0000z.nc4
- Creating file for Restart; reference = 20190701_010000
  with filename = ./Restarts/GEOSChem.Restart.20190701_0100z.nc4
--> DATE: 2019/07/01 UTC: 01:00
- CLEANUP: deallocating arrays now...
```

GEOS-CHEM TIMERS

Timer name	DD-hh:mm:ss.SSS	Total Seconds
GEOS-Chem	: 00-00:03:15.054	195.055
HEMCO	: 00-00:02:05.070	125.070
All chemistry	: 00-00:00:09.460	9.461
=> Gas-phase chem	: 00-00:00:05.804	5.805
=> Photolysis	: 00-00:00:01.015	1.016
=> Aerosol chem	: 00-00:00:02.195	2.195
=> Linearized chem	: 00-00:00:00.039	0.039
Transport	: 00-00:00:01.554	1.555
Convection	: 00-00:00:00.960	0.961
Boundary layer mixing	: 00-00:00:01.296	1.297
Dry deposition	: 00-00:00:00.140	0.141
Wet deposition	: 00-00:00:01.117	1.117
Diagnostics	: 00-00:00:01.242	1.242
Unit conversions	: 00-00:00:01.046	1.047

```
~~~~~
% To compute the OH metrics, execute the Python
% script "metrics.py" in this run directory.
%~~~~~
```

Checking if the variable exists:

```
ncdump -h GEOSChem.ProdLoss.20190701_0000z.nc4 | grep -E
```

```
'Prod_HNO3other|Prod_HNO3N2O5H2O|Prod_HNO3NO2OH'
```

```
[b169195@chpc-sandbox OutputDir]$ ncdump -h GEOSChem.ProdLoss.20190701_0000z.nc4 | grep -E 'Prod_HNO3other|Prod_HNO3N2O5H2O|Prod_HNO3NO2OH'
```

```
float Prod_HNO3other(time, lev, lat, lon) ;
  Prod_HNO3other:long_name = "Chemical production of HNO3other" ;
  Prod_HNO3other:units = "molec cm-3 s-1" ;
  Prod_HNO3other:averaging_method = "time-averaged" ;
float Prod_HNO3N2O5H2O(time, lev, lat, lon) ;
  Prod_HNO3N2O5H2O:long_name = "Chemical production of HNO3N2O5H2O" ;
  Prod_HNO3N2O5H2O:units = "molec cm-3 s-1" ;
  Prod_HNO3N2O5H2O:averaging_method = "time-averaged" ;
float Prod_HNO3NO2OH(time, lev, lat, lon) ;
  Prod_HNO3NO2OH:long_name = "Chemical production of HNO3NO2OH" ;
  Prod_HNO3NO2OH:units = "molec cm-3 s-1" ;
  Prod_HNO3NO2OH:averaging_method = "time-averaged" ;
```